CFRM 502, Winter 2020.

Homework 1

Due: Wednesday 15 January 2020, NO LATER than 11:59pm

Instructions: You must submit your work through Canvas, and in the form of two attachments: A .pdf file of your detailed answers and a file containing your codes (.R or .Rmd). The .pdf files can be typeset or be a readable scan of your handwritten notes. You do not need to include R outputs in the .pdf file. However, you must clearly indicate the relevant part of your R codes when answering a question that is based on your code.

The maximum number of points you can receive for this homework is 26.

1. Consider the daily prices of Caterpillar stock (CAT), Procter & Gamble stock (PG), and Dow Jones Industrial Average (^DJI) from January 3, 2007 to December 31, 2018.

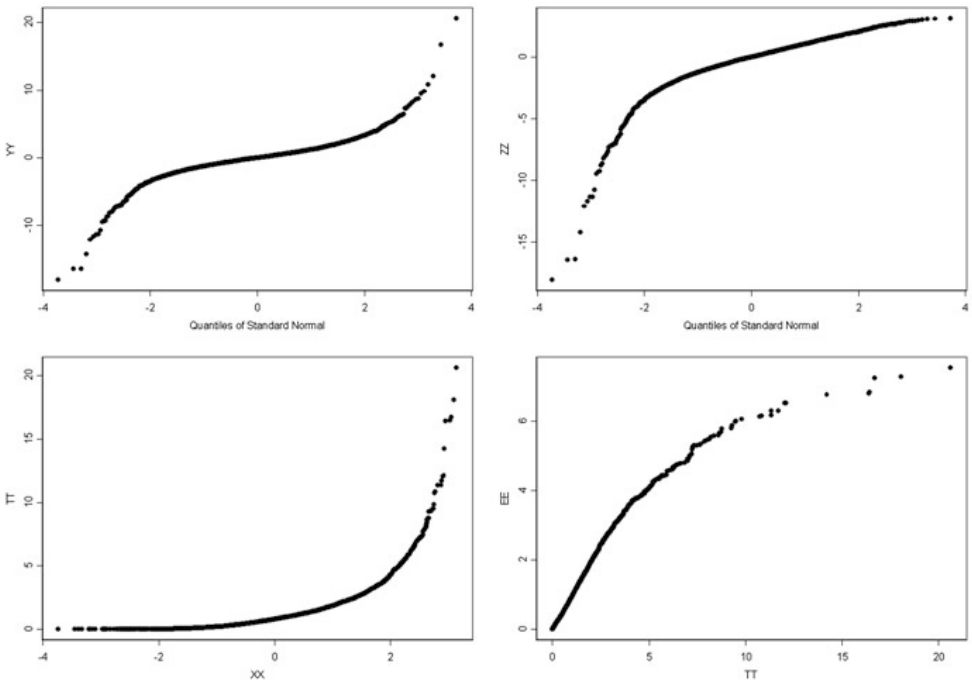
   (a) (1 pts) Provide a candlestick plot for each of the three time series.

   (b) (3 pts) Calculate the net daily returns for the three prices. Find basic statistics for the sample returns, namely the sample mean, standard deviation, minimum and maximum, median, and 1st and 3rd quartiles of each return series.
   Hint: You may find the function "basicStats()" of the R package "fBasics" useful.

   (c) (3 pts) Are the **log returns** of the prices normally distributed? Justify your answer by an appropriate test or graph.

   (d) (3 pts) Find a t-distribution that is a good fit for the marginal distribution of each **log return** series. You must justify your answer by appropriate graphs.

2. (a) (2 pts) Consider 5 samples named EE, TT, XX, YY, and ZZ. The following 4 graphs are Q–Q plots. The top two are normal probability plots of YY and ZZ samples, while the bottom two graphs are the quantiles of two samples. Give an interpretation to each of the Q–Q plots.



(b) (2 pts) Use the R command "rnorm" to create a sample of size N = 100 from the standard normal distribution and use Q–Q plots to assess the normality of the data. In other words, explain which features of the Q–Q plot you produce suggest that the data are from a Gaussian distribution and which don't.

(c) (2 pts) Repeat part (b) for a sample of size N=100 from the exponential distribution (with mean 1). Hint: use the R function "rexp"

(d) (2 pts) Draw a Q–Q plot of the sample in part (b) vs. the sample in part (c). Interpret the Q–Q plot.

3. (a) (4 pts) For the line regression model $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$, find a formula for $\mathrm{Var}(\hat{\beta}_0 | X_1, \ldots, X_n)$. Hint: You may use equation (9.8) on page 222 of the textbook.

(b) (4 pts) Show equation (9.15) on page 228 of the textbook for the line regression model in part (a). That is, show that

$$\sum_{i=1}^{n} \left(Y_i - \overline{Y}\right)^2 = \sum_{i=1}^{n} \left(Y_i - \widehat{Y}_i\right)^2 + \sum_{i=1}^{n} \left(\widehat{Y}_i - \overline{Y}\right)^2$$

where $\overline{Y} = n^{-1} \sum_{i=1}^{n} Y_i$ and $\widehat{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 X_i$.